

Contracting structures in public procurement: Evidence from donor-funded electrification in Kenya

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September 17, 2026

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Abstract

This paper studies how contracting and monitoring structures in public procurement affect large-scale infrastructure construction. Between 2016 and 2022, Kenya’s public electric utility ran competitive auctions and then awarded and administered dozens of donor-funded procurement contracts to connect millions to electricity. The causal analyses leverage natural and experimental policy variation along with granular, independently-collected construction quality data. Despite largely harmonized procedures, Kenya Power contracts funded by the World Bank featured unbundled contract components—known as ‘design-bid-build’—and enhanced monitoring compared to contracts funded by the African Development Bank. The unbundling of construction contracts delayed construction progress but improved quality, while randomized inspections improved quality without large delays. The results suggest a combination of bundled contracting with additional independent monitoring could improve outcomes.

JEL codes: F35, H57, L94, O19

Authors are in ©[Certified Random](#) order. Wolfram: Massachusetts Institute of Technology and NBER. Miguel: University of California, Berkeley and NBER. Hsu: University of California, Berkeley. Berkouwer: University of Pennsylvania and NBER. We thank the Foreign, Commonwealth and Development Office (FCDO), the Kleinman Center, and Analytics At Wharton for generous financial support. We thank Baba Fatajo, Benjamin Olken, Zubair Sadeque, Duncan Thomas, Giulia Zane, and seminar participants for helpful

1 Introduction

Government agencies often rely on private firms to supply goods and services: public procurement spending amounts to 12% of global GDP (Bosio et al., [2022](#)). In low- and middle-income countries (LMICs), contracts between governments and private firms often receive financing from multilateral institutions. Between 2000 and 2022 the World Bank financed more than 311,000 government contracts for the procurement of more than \$185 billion in goods and services for over 21,000 projects across more than 170 countries. But procuring public goods from private firms may misalign incentives, complicating a government’s ability to provide high-quality infrastructure projects at reasonable cost. Contracting practices can improve project outcomes but may also introduce bureaucratic inefficiencies or inhibit useful contractor discretion (Liscow, Nober, and Slattery, [2023](#); Williamson, [1999](#); Hart, Shleifer, and Vishny, [1997](#); Bosio et al., [2022](#)).

Despite the importance of this problem, causal empirical evidence on the impacts of contracting practices is scarce. Because infrastructure projects tend to be large and subject to government regulations that apply nationwide, procurement procedures usually cannot be randomized. Existing sub-national variation in contracting regulations may be endogenous to economic, geographic, or political factors that affect the selection and performance of contractors. Much of the empirical work studying contracting focuses on auctions and bid comments. We thank Kenya Power for generously sharing administrative data. Carolyn Nekesa, Jane Adungo, and Joseph Otieno superbly implemented field activities. We thank Jensen Fiskin, Andrew Hinchberger, Oliver Kim, Robert Pickmans, Nachiket Shah, Adam Streff, Matthew Suandi, Kamen Velichkov, Felipe Vial, Aidan Wang, Katie Wright, and Liyuan Yang for excellent research assistance and nLine for their support deploying the GridWatch technology. A pre-analysis plan was registered with the AEA RCT Registry ([ID 2389](#)). This project received IRB approval in Kenya (Maseno MSU/DRPC/MUERC/27/13) and the U.S. (U.C. Berkeley CPHS 2016-11-9365). On-line appendix [here](#).

evaluation rather than on contracting structure.

This paper’s main contribution is empirically studying how contract bundling and monitoring affect the design, supply, and installation components of a major real-world infrastructure project in a high-stakes public procurement context. An additional contribution is to monitor the execution of publicly procured infrastructure using independently collected on-the-ground construction quality data. We use policy and experimental variation to generate causally identified evidence on this topic in the context of the Last Mile Connectivity Project (LMCP), one of Kenya’s largest recent public infrastructure projects. In 2016 the Government of Kenya (GoK) selected 7,492 villages where all unconnected households within 600 meters of the existing grid would be connected to electricity. The program was implemented by Kenya’s state-controlled electric utility, Kenya Power, which administered competitive auctions and awarded dozens of contracts to private firms to conduct construction activities.

LMCP construction was funded by the World Bank (WB) and the African Development Bank (AfDB). Multilateral financial institutions (MFIs) such as the WB and the AfDB typically do not carry out procurement themselves. Rather, public agencies in the borrower country implement projects largely independently, following the procurement procedures mandated by the funders. In the case of the LMCP, Kenya Power staff (representing the GoK) independently managed all aspects of contracting, including administering auctions, reviewing bids, awarding contracts, and monitoring construction. This structure is being used widely: in early 2025, the WB and the AfDB launched ‘Mission 300’, a \$48 billion initiative to connect 300 million Africans to electricity by 2030 (AfDB, [2025](#)).

Most MFIs have largely harmonized procurement procedures to consensus best-practices. To quantify contracting differences and similarities, we use a large language model to review hundreds of pages of documentation, including procurement guidelines, bid documents, technical specifications, and procurement contracts. The analyses reveal that the bidding documentation, evaluation, and procurement regulations in use by Kenya Power for contracts funded by either donor were overwhelmingly either identically worded, or worded

slightly differently but very close in substance. More than two-thirds of sentences in AfDB procurement regulations can be matched to at least one almost identical sentence in a WB regulation document, and the six distinct sections of the regulations have a 94% overall similarity according to the algorithm. This partly results from the fact that donors’ institutional procedures have converged as a result of a decade-long targeted effort to simplify compliance. This resulted in the signing of the 2003 Rome Declaration on Harmonisation by 28 aid recipient countries and over 40 development institutions—including WB and AfDB—which aimed “*to harmonise the operational policies, procedures, and practices of our institutions... to improve the effectiveness of development assistance*” (OECD, [2003](#)).

Despite these similarities, we identify two major differences in the procedures that Kenya Power used in implementing the LMCP. The first difference relates to ‘unbundling,’ or how project components are assigned across contracts—a central question in contract structuring. Two of the most commonly used approaches for infrastructure projects worldwide are ‘design-bid-build’, which unbundles design and construction across separate contracts, and ‘design-build’ which bundles these components. For villages funded by the AfDB, Kenya Power awarded ten bundled contracts that included network designs, materials, and installation. In contrast, Kenya Power awarded 29 unbundled contracts for villages funded by the WB.

The second difference relates to the intensity of monitoring for contract outputs. At WB-funded sites, Kenya Power was required to conduct an additional round of inspections before the network was handed over from the contractor to the utility for operation. Best practices in monitoring continue to be debated, with the World Bank’s Independent Evaluation Group noting that “[the] World Bank could have a more consistent process for monitoring whether procured goods, works, and services reach beneficiaries” (WB, [2024](#)).

To identify the causal impacts of contract unbundling and additional monitoring on project outcomes, we leverage a useful program feature: LMCP sites were assigned to be funded by a donor in a way that appears arbitrary and can reasonably be thought of as quasi-random and without obvious regard to factors that would affect project outcomes.

Two key aspects of the setting give us confidence that we can interpret differences between WB- and AfDB-funded sites as due to the contracting differences we highlight. First, 95% of WB-funded sites in the sample are within 10 km of an AfDB-funded site, suggesting little geospatial selection. OLS and machine-learning balance tests indicate sites are largely indistinguishable on observables, including demographic, geographic, satellite, road, census, electoral, and Kenya Power administrative data (still, the econometric analyses include constituency fixed effects to further account for local geographic and socioeconomic heterogeneity). Second, both donors use unbundled and bundled contracting structures in other contexts, and their decision to use two different approaches for the LMCP appears incidental.

The impacts of unbundling and additional monitoring were unclear *ex ante*. During in-person interviews, WB representatives argued that they would improve construction outcomes, while Kenya Power representatives feared that they would lead to administrative costs and delays without delivering substantive benefits. AfDB representatives in turn agreed with Kenya Power that the latter’s extensive experience made the more streamlined procedures (without unbundling or additional monitoring) sufficient to ensure high quality.

To evaluate the impact of strengthened monitoring among bundled contracts, we implemented a randomized auditing intervention designed to mimic Kenya Power’s additional inspections at WB-funded sites. Through in-person meetings, contractors were informed that key aspects of the completed construction at certain sites (randomly selected by the research team) would be measured and reported to Kenya Power, the WB, and the AfDB.

An additional contribution of the paper is the collection of detailed and innovative engineering data across hundreds of villages which evaluate construction progress, low voltage grid quality, metering components, and electricity connection quality. Field teams tracked construction progress over multiple years for 380 LMCP villages through in-person visits and phone calls to village leaders, and then collected three types of on-the-ground data. First, they measured construction quality for key infrastructure components such as electrical transformers, poles, and wires, following Kenya Power engineering standards. Sec-

ond, state-of-the-art sensors deployed with households measured minute-by-minute site-level power outages and voltage quality. Third, field teams conducted socioeconomic surveys to understand connection experiences and energy usage among a representative household sample. These activities build on existing work emphasizing the importance of detailed infrastructure measurement (for example Olken, 2007; Atkin, Khandelwal, and Osman, 2017).

The first main result is that, in terms of costs, construction at sites that saw both unbundling and high monitoring is far slower and leads to fewer pole installations and household connections. By the end of tracking in 2021, 82% of sites without unbundling or high monitoring had seen poles and wiring go up versus only 72% of sites that had both. Among sites where there had been construction, there are 12% fewer poles and 18% fewer customer connections per site at sites with both unbundling and high monitoring. Unbundling and additional monitoring also caused significant delays: household meter activation is completed on average 16 months later. These results do not appear driven by selection of construction firms, which are largely similar on observables.

Second, in terms of benefits, unbundling and additional monitoring improve on-the-ground construction quality by 0.64 standard deviations on average, and 77% of these sites have higher measured quality construction than the median site without unbundling or additional monitoring. Specifically, poles are 23% more likely to have all key attributes of a high quality pole (namely, a pole cap, no cracks, and a correctly installed strut and stay, when required). There are no measured medium-run differences in electricity reliability and voltage quality, though, and the impacts of these joint procedures on household installation quality, cost, and energy usage are positive but modest in size and generally not statistically significant. Overall, improvements in construction quality at sites with unbundling and additional monitoring appear more likely to improve pole longevity and reduce long-term maintenance costs than to improve service quality in the short- to medium-run.

Could quality improvements be achieved without adding delays by enhancing monitoring without unbundling contracts? The randomized audit experiment shows that the audits

have no impact at sites that used unbundled contracting and had already seen additional inspections. On the other hand, the audits significantly improve installation quality at sites that did not have these procedures, where they increase the household installation quality index by 0.23 and the household electricity usage index by 0.28 standard deviations. Households at these sites experience higher power quality as a result: the audit treatment halves the average gap between experienced and nominal voltage. Contractors installed 18% more poles (and 4% more customer connections, though this latter effect is not significant), and treatment households further report higher household connectivity and energy usage. These results indicate that additional monitoring for projects carried out using bundled contracts can generate similar quality improvements as the combination of contract unbundling and additional monitoring, but with fewer delays and at lower cost.

The final section of the paper assesses the trade-off in costs and benefits for contract unbundling and monitoring. The average cost per new household connection is approximately \$563 at AfDB-funded sites and \$728 at WB-funded sites (so 30% higher at sites with unbundling and additional monitoring), driven by lower per-site costs and more new connections at AfDB-funded sites. The net benefits depend on observable project attributes (such as the delay caused by unbundling) as well as unobservable ones, such as the funder's discount rate and time horizon, and the impact of improved construction quality on long-term maintenance costs. Under a plausible range of assumptions, the net benefits could range from -4% to 7% of project costs in favor of bundling and less-intensive monitoring.

These empirical results point to a meaningful intertemporal trade-off. Policymakers may need to evaluate the long-term benefits of unbundled contracting and monitoring against the apparent short-term costs. Planners with a higher time discount rate (or a shorter time horizon), or those implementing projects with compounding benefits, might prefer the timelier construction enabled by a streamlined bundled contracting approach. Conversely, if maintenance costs are expected to rise quickly with poor quality, delays might be worth the improved long-term outcomes. This perspective can also explain why some political

agents, facing electoral or other domestic pressures, may prefer working with donors whose procurement approach enables greater expediency. In our study context, combining contract bundling with enhanced audits delivers improved quality with little additional cost and delay, and might therefore be preferred to both the AfDB and WB approaches to LMCP funding.

Any relatively short- to medium-run analysis, like ours, has limitations. Procurement procedures may generate additional benefits that we cannot measure, such as enhanced institutional capacity. Like most other research projects, we are also unable to directly measure leakage of funds. However, to the extent that increased leakage at AfDB-funded sites would have reduced construction quantity, we find limited evidence of this: if anything, AfDB-funded sites have more connections at lower average cost than WB-funded sites.

In addition to informing the literature on procurement regulations, these findings also contribute to a broader debate about donor conditionality that dates back at least to the ‘Washington Consensus’ era in the 1980s (e.g. Mosley, 1987; Temple, 2010; Archibong, Coulibaly, and Okonjo-Iweala, 2021). Donor conditions increasingly emphasize procedures (rather than policies), though the resulting costs and benefits have been subject to controversy (WB, 2005; Kersting and Kilby, 2016; Kilby, 2013).

Mass electrification programs are widespread in LMICs, but poor construction quality can harm power quality: in some countries, most connected households “reported receiving electricity less than 50% of the time” (Blimpo and Cosgrove-Davies, 2019). Identifying opportunities to improve construction quality—including through specific procurement contracting conditions—may lead to meaningful improvements in economic outcomes.

2 The Kenyan government’s rural electrification program

Rural electrification in Kenya was spurred by domestic government objectives. In 2006 the Kenyan government passed the Energy Act (2006), creating the Rural Electrification Authority (REA) with a mandate to develop “the rural electrification master plan” and to “source additional funds for the rural electrification programme.” REA’s 2008–2012 Strategic

Plan (2008) to connect public facilities across thousands of rural villages to electricity was followed by Kenya’s Draft National Energy Policy (2014), which targeted universal household electricity access by seeking “funding from development partners.”

By May 2015 the Ministry of Energy had secured funding from two multilateral institutions, and Kenya’s President launched the Last Mile Connectivity Project (LMCP) to achieve universal access by 2020, starting from 25% in 2009 (KNBS, 2009). The LMCP was rolled out as a single nationwide project, implemented entirely by Kenya Power under a uniform set of specifications. Kenya Power outsourced construction to private sector contractors. Contracts were awarded and wholly managed by Kenya Power: they issued, reviewed, and awarded contracts; managed contractors and consultants; and conducted inspections independently. While their ambitious goal was not met, nationwide household grid electricity access reached 50% by 2019 (KNBS, 2019).

Around 60,000 electrical transformers across Kenya convert high- and medium voltage power to low voltage (LV) power suitable for use by households. Many transformers are located in rural villages where very few households were connected prior to the LMCP (Lee et al., 2016). Kenya Power, the Ministry of Energy, and members of parliament jointly selected 7,492 such transformers for the LMCP, targeting an equitable regional distribution across Kenya. The objective was to connect all unconnected households located within 600 meters of an LMCP transformer (typically between 20 and 100). This approach of connecting all unconnected households in a village at once—referred to as ‘maximization’—was intended to generate cost efficiencies by leveraging economies of scale.

Eligible households benefited from a reduced electricity connection price of \$150 (down from \$350), as well as the ability to pay in monthly installments with no upfront down-payment. Households also previously had to organize wiring for power sockets and light switches, which could be prohibitively expensive. Kenya Power therefore decided to provide low-income households with a ‘ready-board’, a standard electrical panel that would satisfy basic wiring requirements (Figure A1). Yet the roll-out of ready-boards was not perfect. Of

the 160 households in our survey sample (described below) who were physically connected to the grid but where electricity had never actually flowed, 45% said it was because they had not yet completed their internal wiring. Note that informal or illegal electricity connections are rare in the lower population density rural areas where the LMCP was implemented ([Appendix C](#) provides additional background).

3 Similarities and differences across contracts

Kenya Power, on behalf of the government of Kenya, outsourced the construction of the LMCP to private sector contractors. They developed uniform tender documentation containing detailed technical specifications for the procurement and installation of poles, wire, conductors, fuses, and meters. They released their requests for proposals through standard channels, reaching many contractors they had previously worked with. Kenya Power personnel then administered competitive auctions for construction contracts with domestic and international private-sector firms.

Multilateral financial institutions routinely finance projects that they do not implement themselves. Instead, funds frequently flow to a recipient government, who is then in charge of implementation. Importantly for this paper’s empirical approach, in the case of the LMCP, the Government of Kenya obtained financing from multiple donors. The AfDB provided \$154 million to ‘maximize’ 4,184 LMCP transformers and the WB provided \$127 million in financing for 3,308 transformers (Kenya Power, [2017](#); [2016](#)).¹ Unlike contexts where projects may incur cost overruns, the contracted amounts in the LMCP appear to constitute a maximum cap. Thus, if realized unit costs ended up being higher than anticipated, then contractors would need to adjust by producing fewer units (and/or lower quality units) rather than increasing total project cost. As a result, the primary outcomes of interest in this study

¹ This paper focuses on activities funded by the WB and by Phase I of the AfDB, which we refer to jointly as Phase I of the LMCP. The publicized numbers differ slightly from the number of observations in the Kenya Power data we use.

are construction quantity and quality rather than overall cost.

Panel A of [Figure 1](#) shows that AfDB- and WB-funded sites were similarly distributed across the country. LMCP transformers were assigned to be funded by the WB or the AfDB in a seemingly arbitrary and ad hoc manner, with neighboring villages often being funded by different donors. [Section 5](#) discusses the assignment process in detail.

3.1 Contracting procedure harmonization and similarity

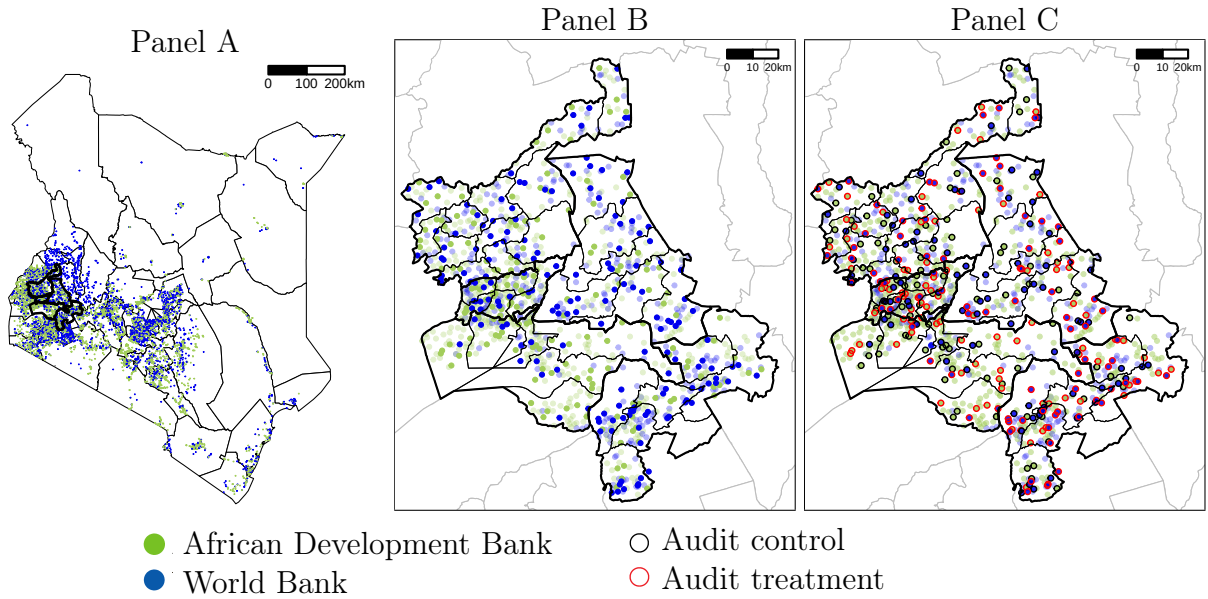
Years of efforts by multilateral institutions to reduce the compliance burden for funding recipients culminated in the 2003 Rome Declaration on Harmonisation, endorsed by “28 aid recipient countries and more than 40 multilateral and bilateral development institutions” (OECD, 2003). WB and AfDB regulations now include harmonized master procurement documents and standard bidding documents ([Subsection C.7](#) offers more detail).

For the LMCP, contracts funded by the WB and AfDB largely used the same standard reverse auction format, employed the same approach for updating bids ex post if reasonable unforeseen changes were required, implemented exceptions for contracts of smaller value (which were selected via quality and cost based scoring), and used the same set of technical household specifications. To quantify the degree of contracting similarity in the context of the LMCP, we review more than 200 documents covering procurement guidelines, bidding documents, contracts, and technical specifications ([Table B1](#)). [Table 1](#) shows that more than two-thirds of sentences in the AfDB procurement regulations match a corresponding sentence in the WB document with a Jaccard score greater than 0.8, meaning they are word-for-word almost identical ([Subsection D.2](#) describes the methodology). The six individual sections have an overall Embedding similarity of 0.94.

Most similar are sections describing the bidding documents, bid opening, evaluation, award of contract, and modified international competitive bidding. Together, these sections determine the auction process and the selection of LMCP contractors.

Our research also included dozens of qualitative interviews with staff at Kenya Power, the

Figure 1: LMCP sites by funding source and audit treatment status



Notes: Panel A shows sites selected for the Last Mile Connectivity Project nationwide. The five counties where we conduct engineering and socioeconomic surveys are marked in bold. Panel B enlarges these five counties and adds within-county constituency boundaries. There appears to be limited spatial clustering by donor. Dark (light) sites are (not) included in our sample. Panel C shows audit treatment and control sites circled in red and black, respectively. Uncircled sites are not in the randomized sample.

Table 1: Text similarity between World Bank and AfDB procurement regulations

Section	Number of sentences in AfDB (2012)	Number of sentences in WB (2014)	(1) Jaccard overlap	(2) Embedding similarity
I. Introduction	55	69	62%	0.90
II. International competitive bidding:				
- General	35	39	63%	0.87
- Bidding documents	108	110	71%	0.99
- Bid opening, evaluation, award of contract	66	67	76%	0.98
- Modified international competitive bidding	16	15	88%	0.98
III. Other methods of procurement	71	92	54%	0.91
Aggregate	351	392	67%	0.94

Notes: Results from a text analysis comparing sections of WB (2014) and AfDB (2012) procurement regulations. Column 1 reports the percentage of sentences in the AfDB procurement regulations that match at least one sentence in the WB regulations with Jaccard similarity greater than 0.8. Column 2 reports Embedding similarity scores, which allow differences in word choice and syntax. [Subsection D.2](#) provides more detail on these methodologies.

WB, and the AfDB, as well as independent construction contractors ([Subsection C.1](#) provides a list of individuals consulted). The information provided across these conversations pointed to the unbundling of contracts and presence of different monitoring activities as far and away the biggest difference between the donors.

3.2 Key difference I: Contract bundling

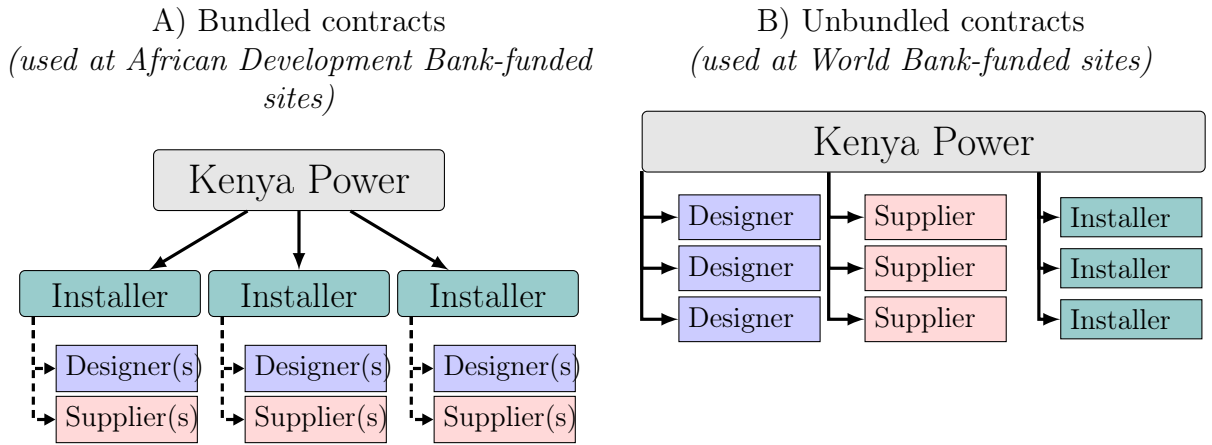
For sites funded by the AfDB, Kenya Power used a bundled contracting approach often referred to in this context as ‘turn-key’ (and sometimes as ‘design-and-build’), which “provides for full design, supply, erection and commissioning of the works by a single contractor at a fixed lump sum price” (AfDB, [2018](#)). Kenya Power auctioned and awarded ten turn-key contracts, each of which comprised the entire construction process of all LMCP sites in one of ten geographical clusters of counties. This process included designing an extension of the LV network to reach unconnected households, procuring necessary materials, and installing these materials. Along with a metering contract and a consulting contract, Kenya Power awarded 12 contracts funded by the AfDB.

For sites funded by the WB, Kenya Power used an unbundled contracting approach. Kenya Power first awarded eight contracts for designs detailing the proposed LV network extensions across eight geographically clustered sets of sites. They then issued 15 contracts to procure materials: six for wooden poles, three for concrete poles, three for conductors, and three for cables. Finally, they issued six contracts for installation at all LMCP sites located in one of six geographic clusters of counties. Kenya Power also awarded two metering contracts and four consulting contracts, for a total of 35 contracts funded by the WB.

[Figure 2](#) shows a stylized framework of these two contracting structures. Bundled contracts are those where the principal only awards installation contracts, and installers are responsible for procuring designs and supplies through subcontracts (Panel A). Unbundled contracts are where each contract contains only one component (Panel B).

These procurement structures are not fixed by donor across all the projects or sectors they

Figure 2: Bundled and unbundled contracting structures



Notes: Schematic of the two types of contracting methods used for the Last Mile Connectivity Project. In the bundled method (Panel A), the principal contracts with installers who procure designs and supplies. In the unbundled method (Panel B), the principal procures components directly. Solid lines represent contracts issued by the principal. Dashed lines represent subcontracts issued by an installer. In the study setting, the principal (Kenya Power) used a bundled structure at African Development Bank sites (awarding 10 bundled contracts plus 2 for consulting and meters) and an unbundled structure at World Bank sites (awarding 29 unbundled contracts for designs, supplies, and installation, plus 6 for consulting and meters).

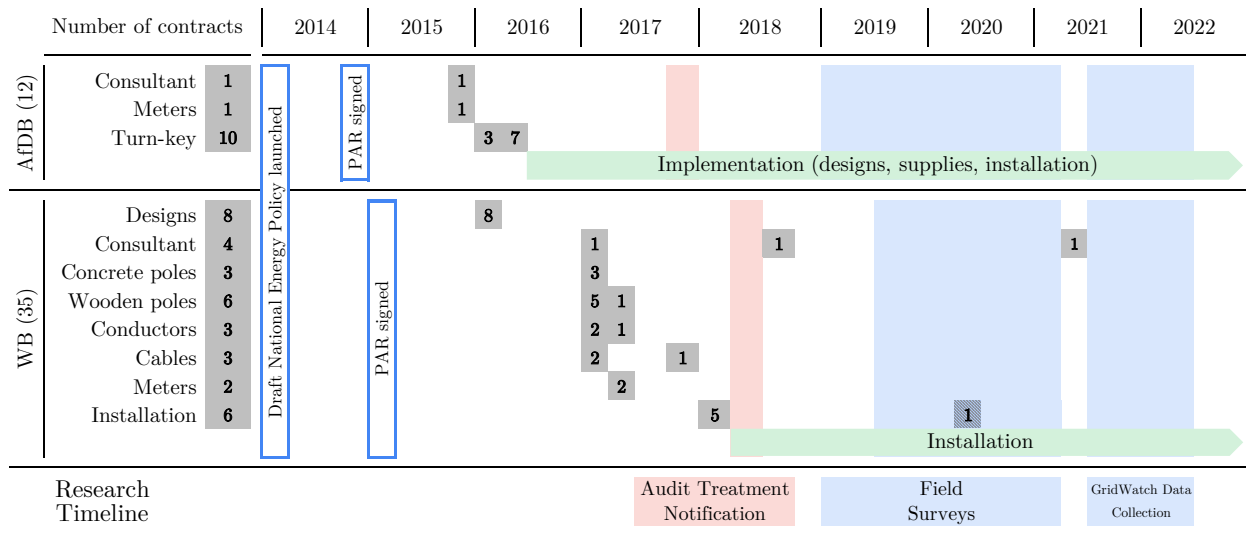
finance globally: the WB often finances bundled contracts, and vice versa. The AfDB and WB decisions to use bundled and unbundled contracting, respectively, for the LMCP were made independently ex ante at the task team level, informed by discussions with Kenya Power and the donors’ experiences in Kenya. WB (2020) states that the “selection of contract types and arrangements takes into account the nature, risk, and complexity of the procurement, and [Value for Money]”. AfDB (2018) similarly states that, “In complex cases, a ‘turnkey’ or ‘design-and-build’ approach may be more appropriate.” Neither funder specifies a strict rule on how this decision is to be taken, but in this case—fortunately for the analysis in this study—they reached opposite conclusions on the most appropriate contracting approach.

Figure 3 presents a timeline of contracting activities. The AfDB and WB Project Appraisal Reports (PARs), which indicate the availability of financing, were signed within five months of each other—in October 2014 and March 2015, respectively—with projected end dates within six months of each other—in December 2019 and June 2020, respectively (AfDB, 2014; WB, 2015). By mid-2016 Kenya Power had signed all 12 AfDB-funded contracts and the eight WB-funded design contracts (Kenya Power, 2015). However, unbundled materials and installation contracts funded by the WB were only processed after designs were completed, and as a result were signed starting February and November 2017, respectively.

3.3 Key difference II: Monitoring and oversight

Oversight in the context of the LMCP can be split into four channels. First, each donor required Kenya Power to conduct similar materials inspections. A team representing Kenya Power’s supply chain, operations, and LMCP departments visited contractors’ facilities to inspect materials. The only difference here was that the WB required each pole to be physically marked to facilitate verification upon arrival at Kenya Power storage facilities. However, this does not appear to have had a large impact: according to Kenya Power staff, spot checks approved over 99% of poles procured through both WB and AfDB contracts. Second, Kenya Power directly monitored each contractor’s activities. Contractors submit-

Figure 3: Dates of contract signing, construction, and research activities



Notes: Timeline of contracting and research activities. The Draft National Energy Policy (2014) spurred government discussions with the African Development Bank and the World Bank, which signed Project Appraisal Reports (PARs) in October 2014 and March 2015, respectively, signaling the official project launches. The gray boxes indicate the number of contracts of each type (in the far left column) and the quarters in which these contracts were signed (in the columns 2014–2022). Sites that had been completed prior to the audit treatment notification in late 2017 were excluded from the randomized sample. Surveys were conducted after construction completion. The date of one consulting contract is unknown.

ted monthly progress reports to Kenya Power staff, who summarized these to share with funders. At least twice per year, each funder conducted similar week-long ‘supervision missions’ consisting of meetings with senior Kenya Power and Ministry of Energy officials in Nairobi and site visits in nearby regions, resulting in a ‘supervision mission report’. Third, the AfDB and WB both required ‘no objection’ approvals at key stages to make sure Kenya Power processes matched the procedures set forth in the funders’ guidelines. Interviews with Kenya Power staff suggest that the WB’s checks were slightly more intensive, but that WB and AfDB checks largely achieved the same compliance goals.

While inspection procedures were largely similar along these first three aspects, procedures differed for the fourth aspect. Specifically, once construction at a site was complete, the consultant, the contractor, and a Kenya Power representative would conduct a joint, on-the-ground inspection of the site and sign a “Joint Measurement Certificate” (JMC) to certify that construction was complete and that the site could be handed over to Kenya Power for activation. However, for WB-funded sites, prior to issuing the JMC, the consultant was required to conduct an additional on-site inspection with the contractor (but without a Kenya Power representative) to produce an “Inspection Report” (IR), listing any observed construction issues or deviations. Comments from the IRs include, for example, “pole caps are poorly installed” and “the strut pole bolt is not secured with nut and washers,” often accompanied by a photograph.

The additional inspection conducted at WB-funded sites could improve quality through two channels. First, the contractor could exert more effort during planning or construction to prevent mistakes from happening, which can be considered a deterrence effect. Second, since IRs were usually conducted ahead of the JMC, contractors could fix remaining issues before the JMC visit. Thus, by identifying issues and incentivizing them to be fixed, the inspections could also have had a direct effect.

The WB and the AfDB use standardized definitions for what behaviors constitute fraud and corruption and thus warrant sanctions. Debarment agreements across donors have for-

malized the practice of agencies recognizing debarments by other agencies. Under-performance can therefore lead to disqualification from contracts by other donors, in other sectors, and even in other countries, making independent audits a meaningful threat.

3.4 Contractor and subcontractor selection

Kenya Power awarded LMCP contracts through competitive processes following standardized rules, including those governing public notification of the bidding opportunity, qualification criteria for bidders, submission of bids, and evaluation of bids ([Subsection C.8](#) offers more detail). The selection process for subcontractors by installers with bundled contracts featured significantly less oversight than Kenya Power’s official auctions. Speculatively, projects with more stringent requirements could attract firms with more efficient operations or better compliance teams. We view this as a mechanism through which procurement regulations aim to improve project outcomes: part of the goal of more stringent regulations is to disincentivize firms who are unable to meet those regulations from submitting bids.

Turn-key contractors often procure designs, materials, and installation from subcontractors. There was some overlap between the contractors awarded WB-funded supplies contracts and the subcontractors from which AfDB-funded turn-key contractors procured goods or services, and the providers listed on bundled and unbundled contracts appear to be similar along observed dimensions. Approximately two-thirds of (sub)contractors were from Kenya, 10% were from China, and 10% were from India ([Figure A2](#)), and these proportions are similar for those awarded bundled or unbundled contracts. 48% of unbundled contractors and 22% of bundled subcontractors had also been awarded at least one other World Bank procurement contract prior to the start of the LMCP. While this could have affected the timing or quality of procured supplies, this does not appear to have been a meaningful issue in practice.

Unbundled contracts could in theory drive down project costs by inducing greater competition at auction, in a setting where more firms have the capabilities to deliver on the

smaller unbundled components. However, project outcomes are suggestive that this was not the case, there being similar numbers of bidders and little indication of lower cost per site for WB-funded construction ([Table B32](#)).

3.5 Theoretical interactions

To offer intuition for how unbundling and monitoring can affect outcomes, consider a principal who wants to contract a project out to one or more firms. Following [Figure 2](#), the project consists of three components: design, materials, and installation. Firms have private types (low or high) and can exert unobservable effort. In the absence of monitoring, bundled contractors choose cost-minimizing design and materials firms and exert minimum effort. Unbundling allows the principal to screen design and materials firms for an eligibility threshold. Meanwhile, additional monitoring incentivizes effort since firms will only be paid if their output meets a quality threshold. In the presence of such a threshold, the screening benefit of unbundling becomes redundant, as lower type firms must expend significant effort to compensate for their low type and (if effort is convex) will not have the lowest bid.

Conditional on being accompanied by high monitoring, bundled contracting should in theory attain similar quality while taking less time ([Subsection D.4](#)). When faced with high monitoring, bundled contractors use their private information and discretion to select design and materials firms to minimize costs while meeting the quality threshold. Under unbundled contracting, firms with the cost-minimizing type and effort for the given quality threshold will offer the cheapest bids. The bundled and unbundled auctions would thus both select optimal firm types and incentivize optimal effort levels for all three components, resulting in identical project quality despite the additional delays caused by unbundling.

4 Data

Enumerators employed by the research team conducted repeated short phone or in-person surveys with village representatives at all 380 sites in the study sample to track construction progress, yielding a site-level panel dataset of construction progress. Reassuringly, nighttime

radiance increases in the 12 months after household electricity metering (Figure A4).

Independently, we conduct on-the-ground engineering assessments and socioeconomic surveys at all 250 sites where construction had made significant progress by the end of the main field activities in May 2021.² 47% of the surveyed sites were tendered through the unbundled WB contracts and 53% through the bundled AfDB contracts (Figure A3 provides an overview of the study samples). Construction had not been completed—and usually not even begun—in the remaining 130 sites, limiting surveying activities there to the short progress assessments. In addition to these independently collected sources of data, we analyze the low-voltage network designs as generated by contractors.

Electricity network data The engineering surveys conducted at these 250 sites were developed in collaboration with recently retired Rural Electrification Authority engineers. In the initial infrastructure census, enumerators recorded the locations and connections of all poles in the low-voltage network up to 700 meters from the central transformer (Figure A5). Only households within 600 meters of the transformer were eligible for an LMCP connection.

Engineering assessments Enumerators recorded characteristics of each pole and the conductors connecting them, focusing on outcomes most likely to affect grid quality and longevity. About a quarter of surveyed poles had a large crack, and 47% were missing a cap (Table B2). At least one fuse was missing or bypassed in around a quarter of transformers surveyed: this could reduce their longevity by exposing them to high-current events.

Household and firm survey data After the infrastructure census, enumerators invited a random subset of connected and unconnected compounds and firms to complete a socioeconomic survey about the construction process and their electricity connections. The survey also asked about manual labor: anecdotally, households are occasionally asked to

² Most sites were surveyed between 4–12 months since metering was completed, which corresponded to on average 12–23 months since the start of construction, with no significant differences between the two funders (p -val= 0.57 and p -val= 0.38, respectively).

contribute manual labor to construction, even though this is strictly against Kenya Power policy. Informal field observations indicate that Kenya Power occasionally installs multiple meters within a single home compound, overstating the total number of households that are connected nationwide (perhaps to inflate program progress statistics). To disentangle this phenomenon from compound residents’ genuine preference for having multiple electricity meters (for instance, if multiple households share a residential compound), the survey asked not just how many meters were installed but also how many they had requested.

Power quality: outages and voltage Improved construction quality could reduce power outages and increase reliability, which could have tangible benefits for household well-being and firm performance. To measure reliability and voltage we plugged in GridWatch devices (Figure A8), which measure minute-by-minute power state and voltage, with a subset of surveyed households and firms (Klugman et al., 2019, 2021). Between June 2021 and June 2022 we collected these data across 150 sites for two months each (reduced from 250 sites due to COVID-19 challenges), deploying four devices per site at a time.

5 Research Design

5.1 Quasi-random assignment of sites to donors

Each LMCP site was assigned to be constructed through contracts that were financed by either the WB or the AfDB. 265 of Kenya’s 290 electoral constituencies contain at least one LMCP site, of which 210 contain at least one AfDB-funded site and one WB-funded site. While geographic clustering might have generated economies of scale by saving on coordination and transportation costs, qualitative interviews suggest donors wanted to avoid perceptions of political bias that might result from funding work in some parts of the country and not others (project documentation does not mention any justification for this decision).

Of the 7,492 LMCP sites shown in Panel A of Figure 1, 1,099 are located in Kakamega, Kericho, Kisumu, Nandi, and Vihiga counties (magnified in Panels B and C), where we

collected detailed on-the-ground assessments. These five counties comprise 36 constituencies, of which 35 have both WB-funded and AfDB-funded sites. There does not appear to be spatial clustering by donor: 95% of WB-funded sites in this sample are located within 10 km of an AfDB-funded site (and vice versa). We randomly select 380 of these 1,099 sites as the study sample, stratifying selection by constituency and funder to improve statistical power.

If utility staff allocated sites to WB or AfDB funding based on site characteristics that affect the quality or timeliness of construction, this could be a threat to identification. For example, this would be the case if sites were allocated to a specific funder to achieve faster construction or better quality for electoral or economic reasons or personal favor.

To address these concerns, we evaluate two related identification assumptions. First, we examine whether the *nationwide* assignment of LMCP sites to construction contracts funded by either the WB or the AfDB was correlated with site characteristics that influence the outcomes of interest. Second, we examine baseline balance among sites *within the study sample*: any differences in the outcomes we measure can then be attributed to the contracting structures. We conduct a range of balance checks, including at the site level and at the constituency level; among our study sample and nationwide; using machine learning methods; and through qualitative interviews ([Subsection D.1](#)).

Across all methods, the assignment of LMCP sites to be awarded contracts funded by either the WB or the AfDB appeared arbitrary and uncorrelated with observable characteristics. The variables that would be most likely to have biased site selection by Kenya Power staff or members of parliament—such as electoral outcomes, ethnic composition, and economic activity—are not significantly different across sites. While some variables are statistically different across the two funders, these differences are economically small and we show below that these variables are weakly correlated with the outcomes of interest.

5.2 Randomized audits

To help disentangle the impacts of unbundling and additional inspections, in collaboration with Kenya Power and the two funders we implemented a randomized audit treatment at a subset of LMCP sites. The same audit treatment was implemented at a random subset of AfDB-funded sites and at a random subset of WB-funded sites. The randomized audits we implemented closely mirrored the Inspection Reports that Kenya Power completed at sites whose contracts were financed by the WB. Similar to those inspections, contractors were informed prior to construction that these audits would take place.

During in-person meetings set up for this purpose, senior Kenyan research personnel notified contractors that an independent, international team of engineers would audit specific sites once construction was complete. They provided a written notice to this effect ([Figure A11](#)), signed by senior management at Kenya Power, the WB, and the AfDB. The notice explicitly listed a set of evaluated outcomes, including distance between poles, line sag, the quality of the connection between the transformer and the low-voltage wiring, and blackouts and reliability. This list emphasized the household experience with electrification more than the standard WB inspections, which focus more on the electricity network itself.

In addition, the research personnel shared a list of sites in their contract region that would be audited. Unbeknownst to the contractor, this list of sites was a randomly selected subset of the full set of sites where our research team conducted endline engineering surveys. Of the 380 study sites, we randomly assigned 190 to treatment and 190 to control, stratifying by constituency and funder. [Figure 3](#) displays the timeline for audit treatment notification and engineering surveys. Panel C of [Figure 1](#) maps treatment and funder assignments. After construction at a site was completed, enumerators hired by the research team visited each site to inspect the electricity network according to specifications we developed in collaboration with retired government engineers. While it is possible that some contractors (correctly) believed that control sites might also be audited, such spillovers are likely small. On average, only 7.6% of a contractor’s sites were audited, and treatment effects do not vary meaningfully

by what share of a contractor’s sites were audited ([Table B10](#)).

Since the audit treatment doubled as the endline data collection exercise, there was no time between the audit and the final measurement, and the contractors thus did not have an opportunity to fix any mistakes identified during the audit. The randomized audit treatment thus isolates a deterrence channel (in which the contractor exerts effort to minimize mistakes) rather than a response channel (in which the contractor can exert additional effort in response to audits to correct mistakes).

In communications with officials, the WB indicated they would take contractor-level outcomes at both WB- and AfDB-funded sites into account in future contracting. This setup can therefore be thought of as a repeated game environment where there are real consequences to contractor performance: many contractors depend on their ongoing relationships with international donors.

5.3 Treatment interactions and inference

The interaction of experimental and natural policy variation allows us to empirically investigate how unbundling and monitoring may interact. We use the following regression specification:

$$y_i = \beta_0 + \beta_1 UM_i + \beta_2 \text{Treat}_i \cdot UM_i + \beta_3 \text{Treat}_i \cdot B_i + \mathbf{\Gamma}_i + \epsilon_i, \quad (1)$$

where UM_i indicates whether site i was assigned unbundled contracting with additional monitoring (observed at WB-funded sites), B_i indicates whether site i was assigned bundled contracting with no additional monitoring (observed at AfDB-funded sites), and Treat_i indicates whether the site is an audit treatment site. Comparing WB-funded sites with AfDB-funded sites among audit control sites allows us to estimate the joint effect of unbundling contracts and enhanced monitoring—this is represented by β_1 . Comparing audit control and treatment sites among the group of sites with unbundled contracting and additional monitoring (observed at WB-funded sites) allows us to estimate the effect of enhanced monitoring in a setting with (already) intensive monitoring and unbundled contracting—this

is represented by β_2 . Comparing bundled contracting sites in the audit control with bundled contracting sites in the audit treatment (observed at AfDB-funded sites) allows us to directly estimate the effect of additional monitoring in a setting with low baseline monitoring and bundled contracting—this is represented by β_3 . Should there be any other procedures affecting project outcomes that differ between WB- and AfDB-funded sites (discussed in [Section 3](#)), or should there be baseline differences between WB- and AfDB-funded sites affecting project outcomes (discussed in [Subsection 5.1](#)), this would be captured by β_1 . We may also compare sites with bundled contracting in the audit treatment against sites with unbundled contracting and enhanced monitoring. To the extent that the audit treatment mimics the enhanced monitoring for WB-funded sites, $\beta_1 - \beta_3$ represents the impact of unbundling (in the presence of enhanced monitoring).

Γ_i is a vector of fixed effects (including constituency fixed effects) which vary somewhat across specifications as discussed below and in the table notes. Standard errors are clustered by site in all regressions except those run at the site level.

If the monitoring at WB-funded sites had the same impacts as our experimental monitoring, the effect of the WB’s unbundled contracting per se can be recovered by subtracting the audit treatment effect among AfDB-funded sites from the aggregate WB versus AfDB difference among audit control sites (i.e., $\beta_1 - \beta_3$ in the regression below). However, we interpret this calculation with caution, as there are modest differences between the inspections Kenya Power implemented at WB-funded sites and the audits our research team conducted.

6 Results

We identify three main results. First, while additional audits alone caused modest delays, sites that also fell under unbundled contracting saw delays that were significantly more severe. Second, joint unbundling and additional monitoring causes considerably higher quality of construction. Third, audit treatments improve the quality of construction at sites without unbundling or monitoring but not at sites that already have these procedures in place. This

indicates that unbundling contracts and monitoring may be substitutes, for example because bundled contractors had more discretion in changing site designs or supplies in response to the audit treatment, whereas unbundled installers were constrained by their assigned designs and previously procured supplies. Alternatively, the audit treatment may have had a limited impact at WB-funded sites due to the diminishing impact of enhanced inspections, above and beyond the additional inspections that Kenya Power had already carried out.

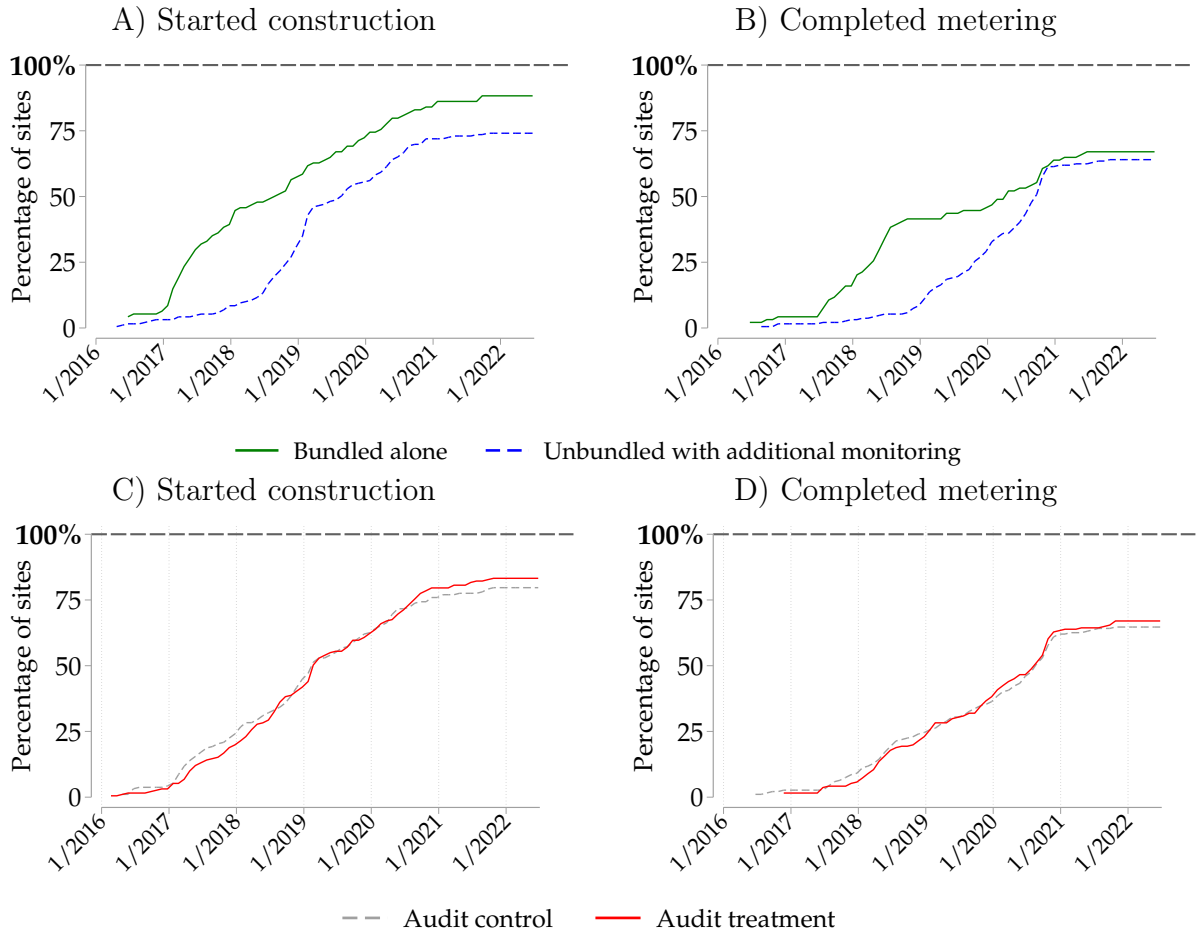
6.1 Construction timing and site completion

Of the 380 LMCP sites tracked by the survey team, 250 saw significant pole construction by the end of field surveying in May 2021. Construction varies significantly by contracting structure: pole construction and wiring had been completed at 72% of sites with unbundled contracting and additional monitoring, but at 82% of sites without these procedures. This difference remains significant even when including constituency fixed effects. In terms of confounding variables, sites with higher baseline nighttime radiance or with a higher land gradient were less likely to be completed ([Table B11](#)), though not differentially so across funders ([Table B12](#)). Still, the regressions presented below control for land gradient.

Even conditional on completion, construction progress at sites with both procedures lagged significantly behind. Panel A of [Figure 4](#) shows that the lag stems from the initial delay in starting construction, likely driven by the ex ante administrative burden involved with contract unbundling. Construction at sites with unbundling and monitoring started on average 10.2 months later ([Table B13](#)): in mid-2018, as construction at these sites was just beginning, the remaining sites reached 50% metering completion. This is in line with Kenya Power’s own nationwide progress metrics (Kenya Power, [2018](#)).

However, once construction started, it proceeded more quickly at sites with unbundled contracts than at sites with bundled contracts, possibly because by that time all designs had been completed and materials supplied. The delay in stringing completion is therefore slightly less, at 9.5 months. However, the delay is then again exacerbated at the final house-

Figure 4: Construction progress by contract structure



Notes: Construction progress data for sites located in the five study counties collected through phone surveys with village representatives. Panels A and B separate progress by whether sites were constructed under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites). Panels C and D separate progress by audit treatment status. [Figure A12](#) displays progress for pole installation and stringing. [Figure A13](#) displays cumulative and conditional progress.

hold metering stage, where the average lag is 16 months. At AfDB-funded sites, meters for example were not always checked to confirm they were functioning prior to issuing a JMC. The more stringent WB inspection reports, which happened between stringing completion and metering activation, may explain the larger delays at this final stage (Figure A14 plots conditional completion times). It is worth noting that, since the sites with unbundled contracts were constructed later, Kenya Power may have experienced some learning from earlier-constructed sites. If anything, this should have led to construction being completed faster at these sites, which is the opposite of what we find.

All sites lagged significantly behind the initial contracting timelines: project appraisal documents developed by both funders originally planned for construction to be completed by early 2019. As seen in Figure 4, lagged progress at sites with unbundled contracting and additional monitoring was apparent even before March 2020, suggesting that COVID-related disruptions played little part in creating these differences.

The audit treatment caused some delays, but these are substantially smaller than the average delay at sites that also had unbundled contracts: metering is completed on average 4.7 months later at audit treatment sites than at audit control sites. To isolate the impact of the unbundling directly, we estimate the impact of joint unbundling and additional monitoring (observed at WB-funded sites) and subtract the audit treatment effect ($\beta_1 - \beta_3$ from equation 1). This analysis indicates that unbundling per se caused a delay of 11 months ($p\text{-val} < 0.001$; Table B13 presents the full regression results). The randomized audit treatment delayed construction start by 4 months but did not cause any additional delays in later stages of construction. In addition, the randomized audits operated exclusively through a deterrent effect. It therefore seems plausible that the delays during the final stages of construction might have been caused by the administrative burden of completing an inspection, processing the results and compiling an inspection report, notifying the contractor, and then after the contractor makes any changes, planning the JMC visit.

6.2 Quantity of construction

Household access to electricity requires the construction of poles to carry electricity throughout the LV network, as well as customer connection cables to connect households to these LV wires. [Table 2](#) shows that sites with unbundled contracts and monitoring saw fewer poles and fewer customer connections (the equivalent regression coefficients from [Equation 1](#) are marked β_1 , β_2 , and β_3). There are on average 81 poles at these sites compared with 92 poles at sites without these procedures, and 58 and 72 new customer connections, respectively. There are several potential explanations for these differences, but one likely explanation is that installers at unbundled sites might have been constrained by the quantity of materials that had earlier been purchased through the separate supply contracts, whereas bundled contractors could procure additional materials as needed during the installation phase.

Column (2) of [Table 2](#) shows that the audit treatment increased the number of poles constructed at sites that did not already have additional monitoring but not at sites that did. Columns (5) through (8) indicate that there was some construction between 600 and 700 meters from the transformer, despite the official guidelines indicating that construction was supposed to have extended only up to 600 meters. Sites with unbundled contracts and additional monitoring saw significantly less construction outside the boundary, possibly due to more stringent adherence to official LMCP rules. This could be viewed as a positive outcome (especially if some of these connections are made in exchange for bribes), but does contribute to fewer connections per site. That said, the household survey data indicate similar rates of requests for informal side payments—about 8%—for households and firms inside versus outside the 600 meter boundary. Voltage decreases with distance from the transformer, as expected, but this decrease is uncorrelated with the funder ([Subsection D.5](#)). Isolating the impact of contract unbundling directly as before ($\beta_1 - \beta_3$), unbundling decreased the number of poles by 18.5 ($p\text{-val} = 0.06$) and the number of connections by 22 ($p\text{-val} = 0.04$).

At the end of survey data collection in July 2022, more than five years after the start of

Table 2: Connections and poles installed per site

	Entire site				Outside 600m boundary			
	Poles		Meters		Poles		Meters	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β_1 : Unbundled with additional monitoring	-11.9** (5.9)	-2.2 (10.1)	-12.8** (6.2)	-19.3* (10.7)	-2.2*** (0.7)	-1.1 (1.1)	-1.4*** (0.5)	-0.9 (0.9)
Audit treatment	6.3 (5.8)		4.9 (6.1)		-0.0 (0.6)		-0.1 (0.5)	
β_2 : Audit treatment (unbundled sites)		-3.3 (8.5)		6.8 (9.0)		-0.7 (0.9)		-0.6 (0.7)
β_3 : Audit treatment (bundled sites)		16.3* (8.3)		2.8 (8.8)		0.8 (0.9)		0.6 (0.7)
Observations	250	250	250	250	244	244	244	244
Control Mean	92.26	92.26	72.25	72.25	3.65	3.65	2.85	2.85
F-test $\beta_2 = \beta_3$ (p-val)		0.10		0.75		0.24		0.24

Notes: β_1 , β_2 , and β_3 are estimated as per [Equation 1](#) to estimate the effects of construction under bundled contracting without additional monitoring (observed at African Development Bank sites) and under unbundled contracting with additional monitoring (observed at World Bank sites). All regressions include constituency fixed effects. Standard errors shown in parentheses. The sample size in columns 5–8 is slightly lower due to logistical constraints while surveying. At the dropped sites, we can calculate the quantities of poles and connections using the engineering survey, but we lack the GPS coordinates needed to calculate whether they are outside the boundary. * ≤ 0.10 , ** ≤ 0.05 , *** ≤ 0.01 .

contracting, household metering had been completed at 67% of sites with unbundled contracting and additional monitoring and at 68% of sites that had neither (Figure 4 shows construction progress over time). However, this hides significant heterogeneity in partial completion: construction had started at 77% of sites with unbundled contracting and additional monitoring but at 90% of sites that had neither. Thus, among sites where construction had started, construction had been completed at 87% of sites with unbundled contracting and additional monitoring but at only 75% of sites that had neither: construction had been only partially completed at 25% of these sites (Figure 4 shows progress across construction stages in more detail). The large share of partially completed public projects in contexts with limited resources and administrative capacity is in line with previous evidence from low- and middle-income countries (Williams, 2017; Rasul and Rogger, 2018). If these sites are not eventually completed, it suggests notable wasted expenditure on construction with limited or no functionality, with a substantially higher share in AfDB-funded sites.

Despite the LMCP mandate to connect all households and firms within 600 meters of the transformer, 10% of households in each village where construction was completed did not have a physical electricity connection, and 13% of households with a physical connection had never seen electricity actually flow (Subsection C.5 offers more detail).

6.3 Power outages and voltage quality

The GridWatch devices recorded an average of 61 minutes of power outage per day, a substantial amount by any standard. Users also experience imperfect voltage quality: Kenya’s nominal voltage is 240V, but voltage in the audit control group averages 233V (3% lower), which could affect day-to-day appliance use and damage appliances over time. Joint unbundling and additional monitoring did not cause statistically or economically meaningful reductions in outages or improvements in voltage quality over the study period (Table B14). Results are similar when estimating daily or monthly coefficients (Figure A18).

The audit treatment had no measurable impacts on power outages or voltage at sites that

already had unbundled contracting and additional monitoring (although point estimates are positive). However, audits had a statistically and economically meaningful effect on voltage quality at sites that did not: the audit treatment there increased average voltage to 238V, which is closer to nominal voltage of 240V than the control mean of 233V.

6.4 Engineering assessment and survey results

Table 3 presents results using primary outcome indices pre-specified in the pre-analysis plan (Berkouwer et al., 2019). Outcomes 1–3 use site-level observations (largely from the engineering assessments) while outcomes 4–9 use respondent-level observations (largely from the household and firm surveys). All indices are standardized to have a mean of zero and a standard deviation of one.

In one of the central results of this study, Column (1) of Table 3 (β_1) shows that overall construction quality (Outcome 1) was on average 0.64 standard deviations higher at sites with both unbundled contracts and additional monitoring (observed at WB-funded sites). This is driven by increased presence of pole caps, struts, and stays on poles (Table B16)—all technical components emphasized in the WB inspection reports. These features are expected to increase the lifetime of the poles—and thus the entire local LV network—over the longer term. Engineering research suggests that capped poles generally experience inner-pole moisture levels between 8–20% whereas uncapped poles experience levels between 30–80%, well above the threshold of 28–30% “considered necessary for fungal attack” (UPRC, 2018).

Columns (2) and (3) of Table 3 estimate the audit treatment effect among sites with both unbundled contracting and additional monitoring (observed at WB-funded sites) and sites that had neither of these contracting features (observed at AfDB-funded sites), respectively (β_2 and β_3). The estimate in Column (3) corresponds to the impact of enhanced monitoring among sites with bundled contracts. Additional audits did not affect outcomes at sites that already had unbundling and additional monitoring. However, at sites that did not, the audits increased household installation quality by 0.23 standard deviations (Outcome 4).

Table 3: Primary engineering and socioeconomic outcomes

	(1)	(2)	(3)	(4)
	β_1 : Unbundled with Additional Monitoring	β_2 : Audit Treatment (unbundled sites)	β_3 : Audit Treatment (bundled sites)	N
Outcome 1: Construction quality index	0.64*** (0.21)	0.10 (0.20)	-0.03 (0.18)	250
Outcome 2: Network size and configuration index	-0.04 (0.16)	0.19 (0.16)	-0.08 (0.18)	244
Outcome 3: Construction timing index	-0.90*** (0.17)	-0.07 (0.16)	-0.29* (0.17)	250
Outcome 4: Household installation quality index	0.05 (0.12)	0.02 (0.11)	0.23* (0.12)	944
Outcome 5: Household cost, experience, bribery index	0.13 (0.12)	0.04 (0.11)	0.10 (0.10)	944
Outcome 6: Reliability and safety index	-0.11 (0.13)	0.03 (0.14)	-0.02 (0.11)	944
Outcome 7: Knowledge index	0.14 (0.10)	-0.00 (0.09)	0.07 (0.10)	944
Outcome 8: Electricity Usage index	0.12 (0.13)	0.11 (0.10)	0.28** (0.13)	944
Outcome 9: Household socioeconomic outcomes index	0.24* (0.12)	-0.00 (0.13)	0.20 (0.13)	944

Notes: β_1 , β_2 , and β_3 are estimated as per [Equation 1](#). Column (1) displays the impact of unbundled contracting and additional monitoring (observed at World Bank sites). Columns (2) and (3) display the audit treatment effect under unbundled contracting with additional monitoring (observed at World Bank sites) and under bundled contracting without additional monitoring (observed at African Development Bank sites), respectively. Each row presents coefficient estimates from a separate regression. Outcome variables are indices constructed from groups of variables standardized to have mean 0 and standard deviation 1. In rows 1–3, observations are sites. In rows 4–9, observations are occupants of connected compounds. All regressions control for site land gradient and public facility type (given some baseline imbalance along these dimensions) and include constituency fixed effects. Standard errors are clustered by site and shown in parentheses. [Table B15](#) reports the version with interaction terms. * ≤ 0.10 , ** $\leq .05$, *** $\leq .01$. The sub-components for each index are presented in [Table B16](#) through [Table B18](#). [Table B24](#) through [Table B26](#) present results on additional secondary outcomes specified in the pre-analysis plan (Berkouwer et al., 2019).

This is driven by earlier meter activation and higher likelihood of having a working meter (Table B17), as well as increased electricity usage of 0.28 standard deviations (Outcome 8). The improvements in voltage quality and household installation quality likely contributed to the increase in household electricity access and usage estimated here (Outcome 8; Table B18). Outcomes 5, 6, and 7 (household cost and experience; reliability and safety; and knowledge) show little difference by contract structure (Table B19, Table B20, Table B21). These results do not appear to be driven by the fact that unbundled sites saw construction later such that Kenya Power may have had time to learn more about the construction process (Figure A15). The contracting structure distinction is the only dimension that shows statistically significant treatment effect heterogeneity (Table B27).

Isolating the impact of contract unbundling as above ($\beta_1 - \beta_3$) yields an increase of 0.67 standard deviations ($p\text{-val} = 0.001$) on the construction quality index and a -0.61 effect ($p\text{-val} < 0.001$) on the timing index. This suggests that unbundling contracts has substantial impacts on multiple important project outcomes.

The fact that the audit treatment differed somewhat from the WB inspection reports does not appear to drive these results. First, the audit treatment was delivered after contracts had already been awarded, while contractors would have known about the inspection reports before bidding; unlike the inspection reports, the audit treatment thus could not have operated through a selection effect. In practice, however, the contractors and subcontractors responsible for sites using bundled and unbundled contracting are largely similar on observables (Subsection 3.4), suggesting that any selection effects may be limited. Second, the inspection reports investigated more technical components of LV network construction whereas the audit treatment also emphasized the quality of household connections. This could explain why the inspection reports affected core engineering components (Outcome 1) whereas the audit treatment primarily improved household installation quality (Outcomes 4 and 8) and voltage quality at sites without inspection reports.

In some contexts, bundling might incentivize excessively frugal designs so that the in-

stalling firm can save on supply and installation costs. We do not find evidence of this in our context, in the aggregate (Outcome 2 of [Table 3](#)) or looking at specific design features individually ([Table B22](#)). We find no other differences in the underlying designs ([Table B23](#)).

6.5 Robustness

All results in [Table 3](#) control for land gradient and facility type, but not doing so does not qualitatively affect the results. Construction outcomes are generally not correlated with land gradient ([Figure A17](#) and [Figure A16](#)) or with the type of public facility that was initially connected under Kenya’s 2008-2012 rural electrification program ([Table B4](#), [Table B28](#), and [Table B29](#)). We also omit an ambiguous ready-board question in the survey ([Table B17](#)) and exclude one particular contractor that experienced unusual financial circumstances and a legal case ([Table B30](#)), and drop sites that are located within 1,200 meters of another site ([Table B31](#)). None of these adjustments qualitatively affects the results described above. [Subsection D.3](#) presents additional robustness tests. We also explore heterogeneity in the time between construction and power measurement ([Subsection D.5](#)).

7 Cost effectiveness

One of the WB’s internal reasons for choosing unbundled contracting was the belief that it would generate cost efficiencies, specifically in that pooling the procurement of major materials into coordinated nationwide contracts would generate purchaser market power and enable them to secure lower prices at auction. This turns out to have been true on paper: the cost per wooden (concrete) pole was 38% (17%) lower in the WB contracts ([Table B32](#)). However, the contract amounts listed in bundled contracts may not reflect true procurement costs: our conversations with implementation contractors suggested that they sometimes shift labor costs onto materials on paper, as these invoices are paid sooner, generating additional liquidity. In contrast to unbundled contracting, where the principal can observe each component’s purchase cost, these practices are not observable to the principal when they are based on subcontractor relationships. This is an example of the opacity that

bundled contracting can create for the principal.

Crucially, aggregate costs per site and per successful connection were substantially higher at WB-funded sites. Kenya Power awarded \$154 million in contracts for AfDB-funded sites and \$127 million in contracts for WB-funded sites.³ This implies an average cost per household connection between \$454–\$602 for contracts for AfDB-funded sites versus \$676–\$781 (about 30% higher) for contracts for WB-funded sites, depending on assumptions around the trajectories of partially completed sites (Table B32). In sum, it does not appear that the WB was able to carry out lower-cost projects overall: in line with the conceptual framework, contract unbundling increased average costs per connection in this setting.

These cost calculations exclude the additional staff time Kenya Power and other agencies spent to meet the WB’s administrative and monitoring requirements. The greater number of contracts and heterogeneity in legal details across contract types required more staff time to draft and issue calls, review bids, and award contracts. Qualitative evidence suggests this cost was substantial (Subsection C.1 provides an anonymized list of interviewees). Unbundled contracting also created coordination frictions, exacerbating administrative costs and delays. These coordination frictions were substantially lower for bundled contracts, where installers could more easily adjust designs or acquire additional supplies at their discretion.

7.1 Cost-benefit: WB and AfDB contracting structures

We first compare sites that have bundled contracts and no additional monitoring (AfDB-funded sites) with sites that have unbundled contracting plus additional monitoring (WB-funded sites). We focus on audit control sites to avoid confounding these differences with the audit treatment’s heterogeneous impacts.

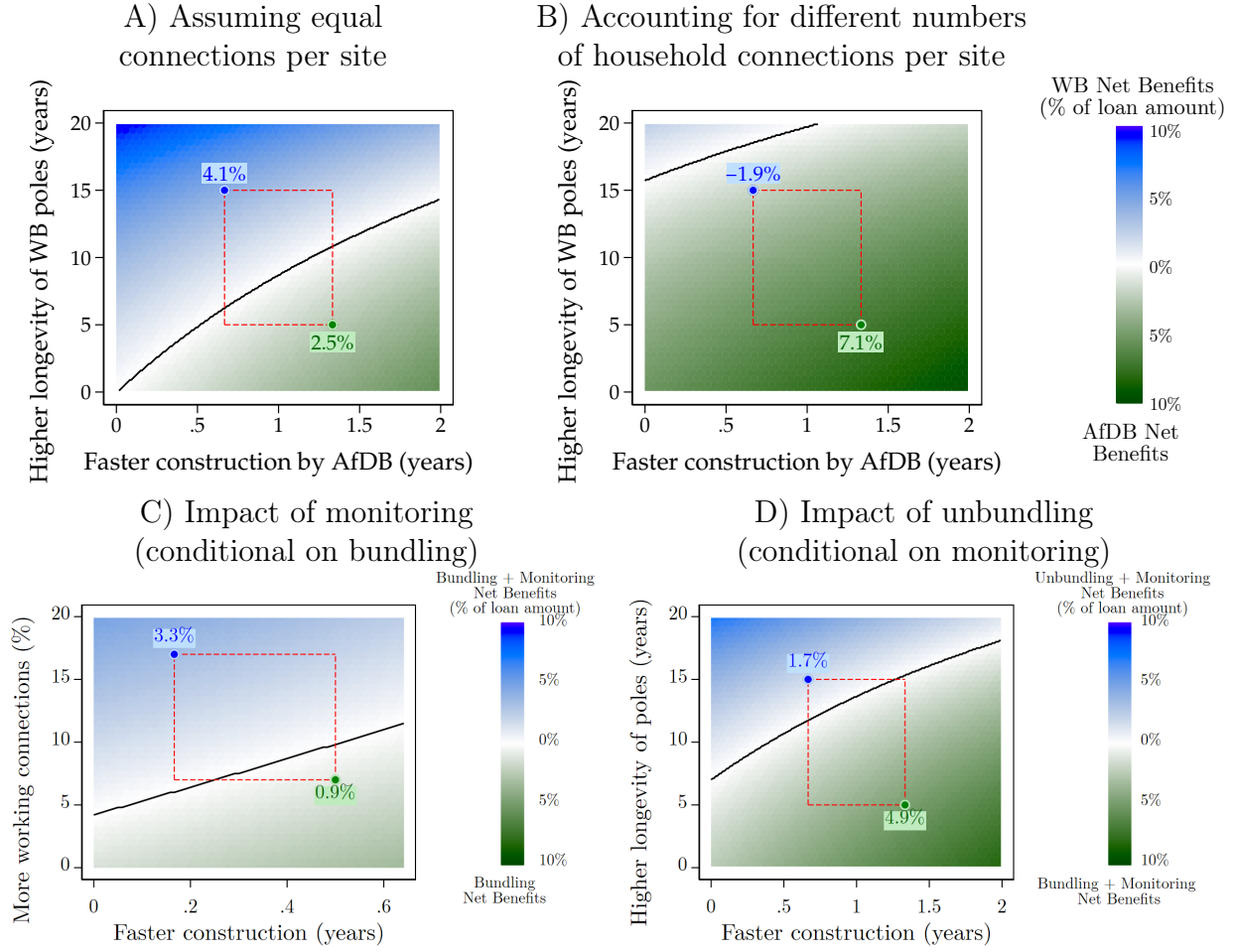
The delays and 30% higher cost per electricity connection for WB-funded contracts documented above might be worth it if gains in construction quality are sufficiently large. We

³ To the best of our knowledge, there were no cost overruns. Rather, increased costs per unit result in a lower quantity of output.

thus evaluate the gains in quality against the cost per connection and the estimated welfare impacts of construction delays to evaluate this key trade-off. AfDB-funded sites reached construction milestones 8 to 16 months earlier than WB-funded sites on average, increasing the net present value of new connections. WB-funded sites saw improved pole and pole installation quality, potentially increasing pole longevity by an estimated 5–15 years and reducing Kenya Power’s long-term maintenance costs (UPRC, 2018). We also factor in that only 71% of sites were completed, and assume that households discount future electricity consumption at a 10% annual discount rate while the social planner discounts future maintenance costs at 5% per year. Figure 5 presents the results in two panels to emphasize the role of one important attribute, namely, the number of household connections. Panel A supposes that WB- and AfDB-funded sites both benefit from 80 new household connections. Panel B reflects our count of LMCP household connections on the ground, which sways the net benefits heavily in favor of AfDB procedures. Using plausible estimates of the gains in construction speed and in quality of poles, the overall net benefits are ambiguous, ranging from a net benefit of 4% of total project costs in favor of WB procedures to a net benefit of 7% of project costs in favor of AfDB procedures. (Figure A19 displays qualitatively similar results under a range of alternative assumptions.)

Under a shorter time horizon, or if the value of a connection to households were larger, the relative benefits of AfDB contracting would be more pronounced, up to 16% of the total. Under a lower discount rate the relative benefits of WB contracting would be more pronounced, up to 5% of the total. However, it appears unlikely that WB procurement procedures would have generated the 30% improvement required to make up for the increased costs per connection: under the most generous assumptions, WB net benefits add up to 5.4% of the loan. These benefit calculations do not consider spillovers such as increased knowledge of oversight mechanisms among Kenyan government agencies, which could positively affect other programs. We also do not consider possible degradation of electricity service quality and reliability over time due to lower quality construction.

Figure 5: Costs versus benefits of different contracting approaches



Notes: Households are assumed to value a connection at \$147 (Lee, Miguel, and Wolfram, 2020) and have an annual discount rate of 10%. The social planner is assumed to have a 20-year time horizon and an annual discount rate of 5%. The red boxes mark reasonable ranges in line with the results above. Panels A and B compare the two donors' contracting approaches. Panel A assumes 80 new household connections per site at all sites. Panel B reflects the observed average 72 household connections at AfDB-funded sites and 58 at World Bank-funded sites. [Figure A19](#) and [Figure A20](#) explore additional assumptions. Panels C and D compare the impact of specific contracting features, assuming additional monitoring costs 500 USD per site. Panel C compares sites with bundled contracting, with and without additional monitoring (observed at AfDB sites in the audit treatment group and the audit control group, respectively). Panel D considers sites with additional monitoring with unbundling (observed at WB sites in the audit treatment group) and without unbundling (observed at AfDB sites in the audit treatment group). The computed net benefits assume that additional monitoring increases the number of working connections per site by 12%, and that non-working connections provide no benefits.

7.2 Cost-benefit: Contract unbundling and additional monitoring

Next, we compare each of these contracting combinations against a third option: bundled contracting with additional monitoring (observed at AfDB-funded sites in the audit treatment group). All calculations assume the additional monitoring costs 500 USD per site (the US\$125,000 budget for field activities across 250 sites surveyed). Electricity flowed to approximately 10% more households, and the audit treatment increased the fraction of households with a working meter by 14% (Table 3, Table B17). Valuing each additional working connection conservatively at \$147, the implied added value of \$1,029 for approximately seven additional working connections per site far exceeds the cost of the audit treatment. This is despite the fact that the audit treatment activity was more involved and thus more expensive than would be needed in other contexts as they also included household and firm surveys, which are not part of the standard inspection reports (IRs) conducted at WB-funded sites.

Panel C shows that additional monitoring among bundled contracts produces positive net benefits in most of the empirically-founded scenarios. Specifically, audits generate significantly more working connections with relatively little cost in terms of delay. Conversely, Panel D shows that if additional monitoring is already in place, then unbundling generates significantly larger delays, which outweigh the modest gain in construction quality that lowers long-term maintenance costs. Furthermore, bundled contracts with additional monitoring tend to compare more favorably to the WB approach than bundled contracts without additional monitoring (Panel A of Figure 5 versus Panel C of Figure 5). Taken together, these results suggest that a combination of bundled contracting with additional monitoring could achieve similar quality gains as unbundled contracts, but with significantly fewer delays.

An important caveat is that we do not directly observe the leakage of funds. It is possible that unbundled contracting meaningfully reduced leakage, which was recently shown to be substantial for WB lending (e.g. Andersen, Johannesen, and Rijkers, 2022). However, to the extent that unbundling increased the availability of funds for construction by reducing leakage, this does not appear to have increased the number of household connections.

8 Conclusion

Outsourcing public goods provision creates a standard misaligned incentives problem: how can a principal identify good contractors and incentivize them to provide high quality projects? Public procurement regulations can thus have important implications for the costs, timeliness, and quality of infrastructure construction, a major spending category for governments and aid donors. One key decision the principal faces is whether to unbundle components of a project across separate contracts auctioned to private firms, or whether to award a single bundled contract that includes multiple components. This decision is ubiquitous in public contracting, but causal inference on this topic has been hampered by the infrequency, endogeneity, and complexity of infrastructure projects.

Unbundling can improve outcomes by enforcing more stringent eligibility criteria on subcontractors, but may add delays and inefficiencies. However, combining bundled contracting with high monitoring may achieve similar quality standards but with fewer administrative costs and delays, because it leverages bundled contractors' ability to manage subcontractors while exploiting synergies inherent in contract bundling.

Comparing the procurement processes in this context generates insights for a trade-off relevant to a wider range of infrastructure projects. A policymaker may need to weigh the short-term benefits of achieving earlier project completion versus the longer-term benefits arising from improved project quality, according to their time preferences. We evaluate this trade-off by comparing the AfDB and WB approaches in the LMCP under a plausible range of assumptions and find that neither approach definitively dominates the other in this context. The results imply anything from a net benefit of 7% of project value in favor of the AfDB approach to a net benefit of 4% of project value in favor of the WB approach.

In this context, enhanced monitoring appears to be an effective and lower-cost substitute for contract unbundling, achieving significant improvements in construction quality without the delays. Unbundling contracts greatly increases the principal's administrative burden,

which—in contexts where staff time and human capital are constrained—can generate substantial delays. The results suggest that combining bundled contracting with more rigorous monitoring could reduce delays while maintaining construction quality standards.

The cost-benefit calculations above illustrate trade-offs in procurement approaches relevant for large-scale infrastructure projects more generally, even beyond rural electrification in Kenya. If the planner discounts future costs and benefits more heavily, if benefits are larger so that delays are costlier, or if unbundling produces especially large delays then bundled contracts may be preferable. For example, when building emergency shelters for disaster relief or clinics in response to a health emergency, the longevity of construction may matter little while delays are particularly harmful. Conversely, if bundling is expected to cause a greater decline in quality—perhaps because it leads to the selection of low quality suppliers—or if remedying low quality construction in the future is especially costly, then unbundled contracts may be preferable. For example, building a bridge or dam may have long-term regional benefits, while any construction defects may be very costly to fix.

Several limitations are worth noting. First, the more stringent WB procurement conditions could generate additional longer term benefits that are hard to measure, such as improved institutional capacity or accounting practices in public sector organizations. Second, while we measure outcomes up to five years after construction, some impacts may only emerge after longer time horizons, including possible differences between WB- and AfDB-funded sites in terms of the longevity of the local grid and power reliability, with gains in WB-funded sites potentially growing over time. Finally, Kenya is a relatively high-capacity state compared to its East African neighbors; its internal regulatory system may be sufficiently rigorous so as not to benefit meaningfully from the additional WB procurement requirements. It is possible that the results would not hold in a setting with weaker institutional capacity, like some of its regional neighbors (as argued for instance by Bosio et al., 2022). Additional research is needed to understand the potentially heterogeneous impacts of procurement processes over time and in other settings.

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